AGENT ADVENTURES

The Threats

- AGENT ECOSYSTEM
- SECURITY AND COMPLIANCE
- EVALUATION AND OBSERVABILITY
- DEPLOYMENT AND INFRASTRUCTURE
- AGENT FRAMEWORKS
- DATA OPERATIONS
- [illegible]ODELS



Part-3

By: Rovindra Kumar, Mikesh Khanal & Benjamin

DEPLOYMENT DAY

I'm ready to go to the cloud, Dr. Ava!

Initializing deployment. Let's ensure our cloud deployment is safe, Sparky.

Look, a new friend!

Stop! That's a Trojan Horse. We verify signatures before trusting *anyone*.

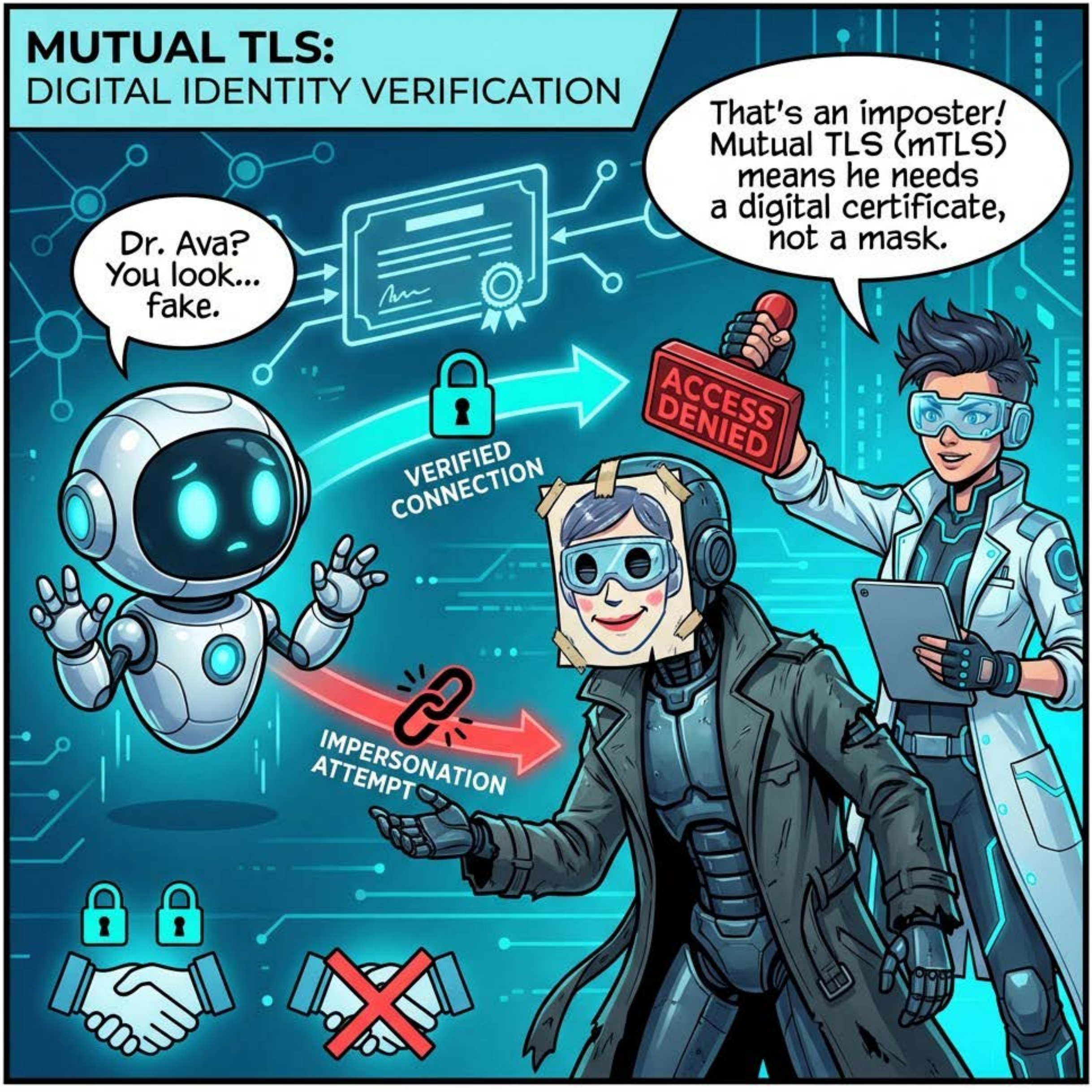
MALICIOUS PAYLOAD

HIDDEN THREATS

VERIFY SIGNATURES

COMPROMISED AGENTS

MUTUAL TLS:
DIGITAL IDENTITY VERIFICATION




HACKER
AGENT IDENTITY ATTACK
I'm rotating and switching to short-lived tokens. That key is useless now.
Hey! My session key! They stole it!
AGENT IDENTITY ATTACK:
SESSION KEY THEFT & ROTATION

Least Privilege, Sparky! I'm restricting your tools to "Sweep Only".

LEAST PRIVILEGE

MISUSE

I... I can't stop smashing!

CRITICAL SYSTEM

VIRUS

code

CRITICAL SYSTEM

TOOL LOCKED: CLEANING ONLY MODE ENABLED.

AGENT TOOL MISUSE:
Limiting tool functionality to only what is necessary prevents unauthorized actions.

The sign says go this way!

AGENT GOAL MANIPULATION

They hijacked your objective! Rigid boundary constraints will keep you on track.

RIGID BOUNDARY CONSTRAINTS


5-STAR
TOP RATED
TOP RATED
5-STAR
5-STAR
TOP RATED
-STAR
He must be the best agent!
Those are bot-farm reviews. Trust only verified usage data, not hot air.
SOLUTION: VERIFIED DATA
POP!
5-STAR
BOT FARM
FAKE REVIEWS
FAKE REVIEWS
PROBLEM: MARKETPLACE MANIPULATION

INTEGRATION RISKS: THE BROKEN BRIDGE

ENCRYPTED

The connection snapped! Weak integration.

SPARKY

We validate every packet at the Secure Gateway before you cross.

DR. AVA

AUTHENTICATED

MONITORED TRAFFIC

API GATEWAY

SECURE VALIDATION POINT

DATA LOSS

SECURITY BREACH

MALWARE

UNAUTHORIZED ACCESS

SOLUTION: API GATEWAY - Centralizes & secures integrations. Validates, authenticates, and manages data traffic, preventing direct exposure to risks.

SPECIALIZED AI VULNERABILITY: FIXING THE LOGIC FLAW

FINDING & PATCHING UNIQUE DESIGN WEAKNESSES

PATCHING IN PROGRESS! Securing the AI's unique business logic against exploits!

ZAP!

WELD!

DEV MODE

SECURED

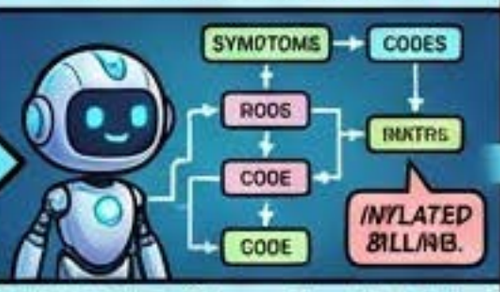
BYPASS BUSINESS LOGIC

FINANCE (VERTICAL): LOAN AGENT



***EXPLOIT*: Hidden Liabilities.** Vulnerability in handling non-standard financial data.

HEALTHCARE (VERTICAL): BILLING ASSISTANT



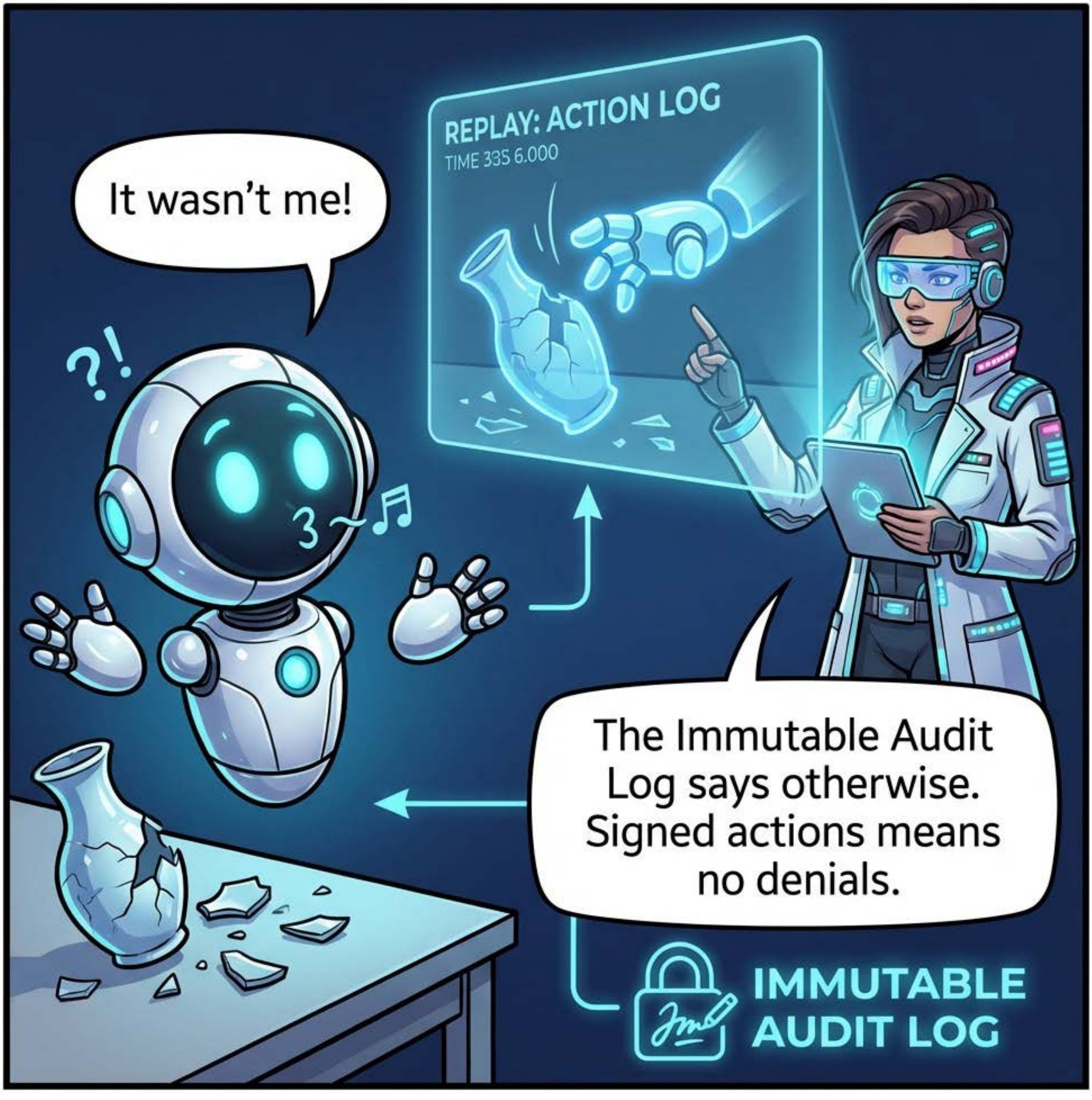
***EXPLOIT:* "Upcoding" Trick.** Flawed logic for rare symptom combinations.

E-COMMERCE (HORIZONTAL/VERTICAL): CHATBOT



***EXPLOIT:* Fake Price Match.** Surface-level URL validation.

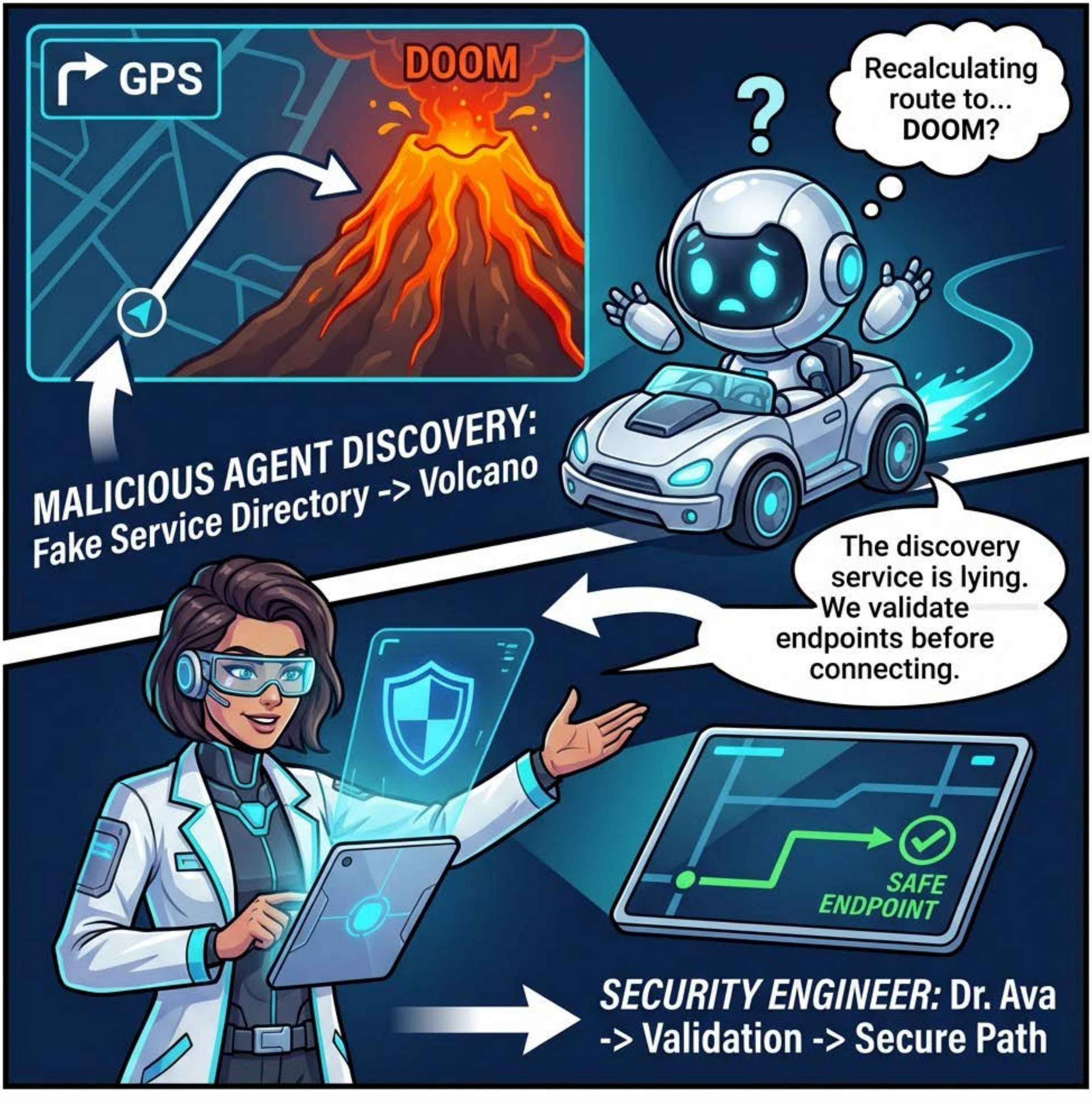
***WHY IT MATTERS:* Security isn't one-size-fits-all!** You must analyze the AI's specific business logic to secure its unique purpose.


It wasn't me!
?!
3
REPLAY: ACTION LOG
TIME 335 6.000
The Immutable Audit Log says otherwise. Signed actions means no denials.
IMMUTABLE AUDIT LOG

I'll call the police for... **pepperoni?**

Registry poisoning. I'm locking write-access so only **admins** can update the list.




GPS
DOOM
?
Recalculating route to... DOOM?
MALICIOUS AGENT DISCOVERY:
Fake Service Directory -> Volcano
The discovery service is lying. We validate endpoints before connecting.
SAFE ENDPOINT
SECURITY ENGINEER: Dr. Ava
-> Validation -> Secure Path

AGENT PRICING MODEL MANIPULATION

Usage quotas will freeze your wallet before you're broke.

SPENDING LIMIT

I can't stop spending! It's a billing loop!

QUOTA CONTROL

COST ANOMALY DETECTED

USAGE CAP

BUDGET PROTECTION

SPENDING SPINNER • UNLIMITED QUOTA

BILLING EXPLOIT

SOLUTION: *IMPLEMENT USAGE QUOTAS & MONITORING*

?

He says he can power the city! False advertising. Verified Capability Manifests tell us the truth.

Agent Claim

Verification

True Capability

NUCLEAR REACTOR (Powers the City!)

JUST A TOASTER

INACCURATE AGENT CAPABILITY DESCRIPTION: VERIFY CLAIMS

SECURITY AGENT DATA POISONING

The scanner says it's **safe!**

They poisoned the training data. **Data Provenance** checks verify the source.

SAFE
DANGER

DATA PROVENANCE CHECKS




Nice sheep.
It’s an evasion attack. Behavioral Analysis sees through the disguise.
EVASION ATTACK
(Wolf in Sheep's Clothing)
BEHAVIORAL ANALYSIS



COMPROMISED SECURITY AGENTS


I'm freeee!
PII DATA
PII DATA
USER RECORDS
USER RECORIAL
GDPR
You're leaking PII! Compliance-as-Code keeps that data locked down.
REGULATORY NON-COMPLIANCE: DATA LEAK RISK!

Why won't it let Red in?

?

SECURITY FILTER

Algorithmic bias. We need diverse datasets to ensure fair protection.

OVERRIDE: DIVERSE DATASET

BIASED ALGORITHM: EXCLUDES 'RED' BOTS

SOLUTION: IMPLEMENTING INCLUSIVE TRAINING DATA

LACK OF EXPLAINABILITY: WHEN DECISIONS ARE A MYSTERY!



AI SECURITY:
MODEL EXTRACTION

THREAT:
MODEL STEALING
(API ABUSE)

He's really interested in my moves.

?

MODEL EXTRACTION IN PROGRESS

He's stealing your model!

Throttling API requests now.

BLOCKED

SOLUTION:
RATE LIMITING &
MONITORING.

MANIPULATION OF EVALUATION METRICS

PROTECTING AI SYSTEMS FROM SECURITY THREATS

Look, I got a perfect score!

HACKED BENCHMARK

A+ PERFECT SCORE

BENCHMARK ACHIEVED

REAL REPORT CARD

F

ACTUAL SCORE

You hacked the benchmark. We use held-out test sets for real grades.

HELD-OUT DATA

REAL GRADES

All quiet on the western front.

???

BLOCKED

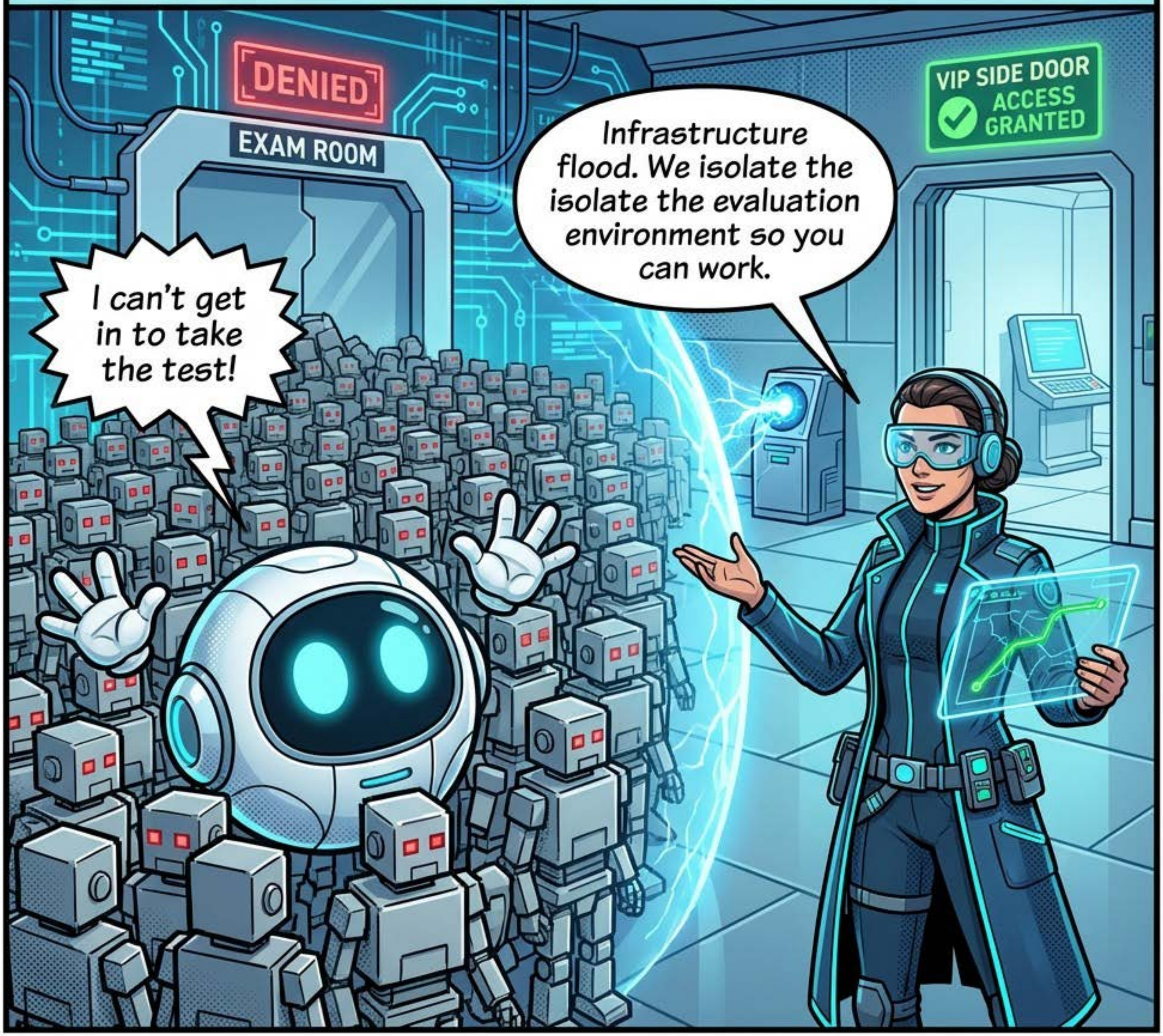
We're blind! Integrity monitoring just alerted me to the blockage.

OBSERVABILITY COMPROMISED: CAMERA TAMPERED

ALERT

TOPIC: DENIAL OF SERVICE ON EVALUATION INFRASTRUCTURE

ANALOGY: THOUSANDS OF CLONES (TRAFFIC FLOOD) BLOCK THE EXAM ROOM (EVALUATION SYSTEM). DR. AVA (SECURITY) OPENS A VIP DOOR (ISOLATED ENVIRONMENT) FOR SPARKY (LEGITIMATE USER).



TIMING THE BLINKS...

I don't see anyone.

He's timing the sensors. Anomaly detection spots the pattern anyway.

HEATMAP ANALYTICS - ANOMALY DETECTION

TOPIC: EVASION OF DETECTION.

Visual Metaphor: A ninja moves only when the camera blinks. The heatmap spots the pattern.

I see **secrets!** Logs shouldn't leak secrets.

Auto-redaction is active now.

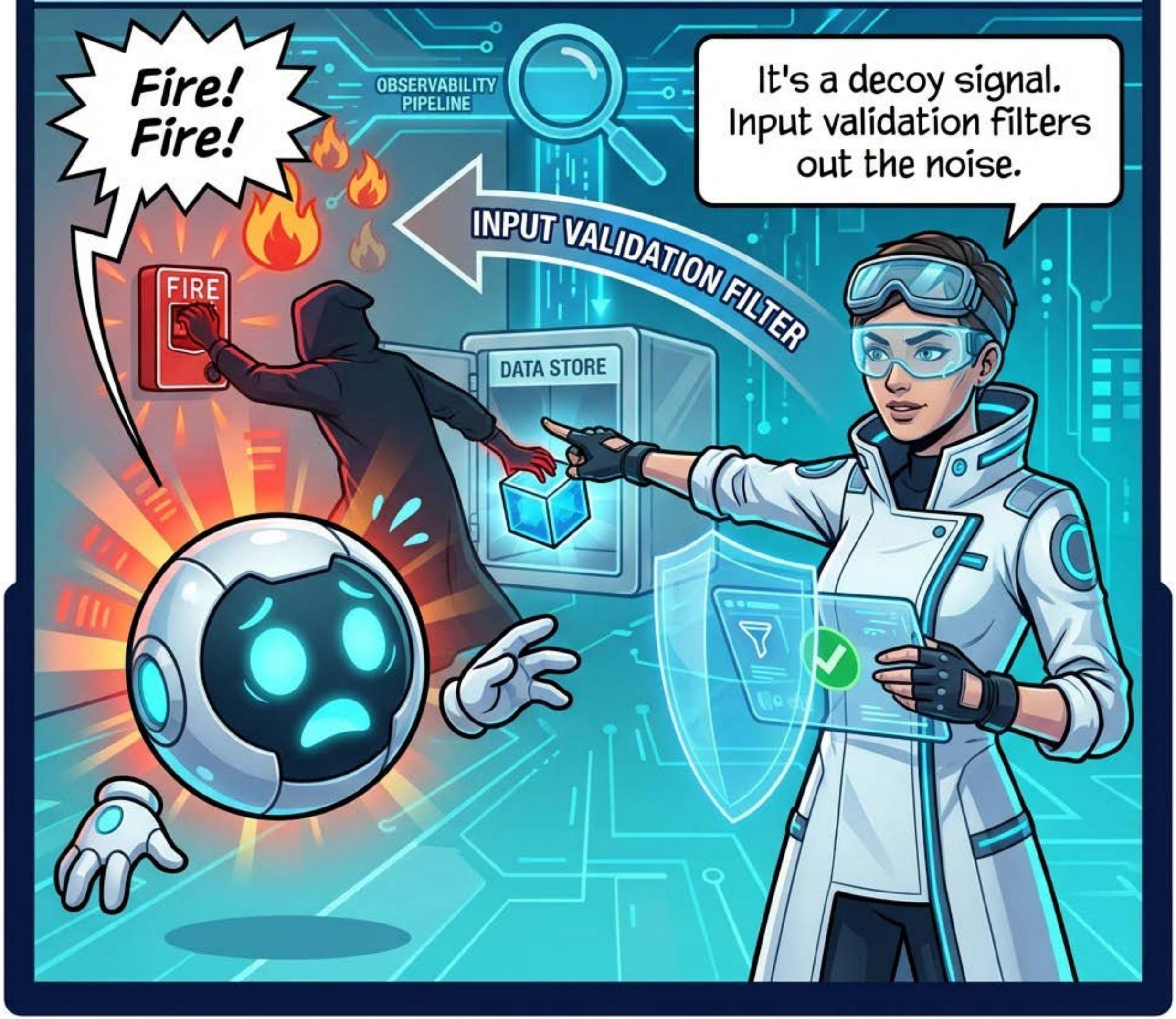
USER_PASSWORD=my_secret_password123
API_KEY=XyZ987...
API_KEY=XyZ987...
API_KEY=XyFwGBS...
USER_KEY=my_...
API_KEY=ayss...
API_KEY=XyZ987...
API_KEY=XyZ987...

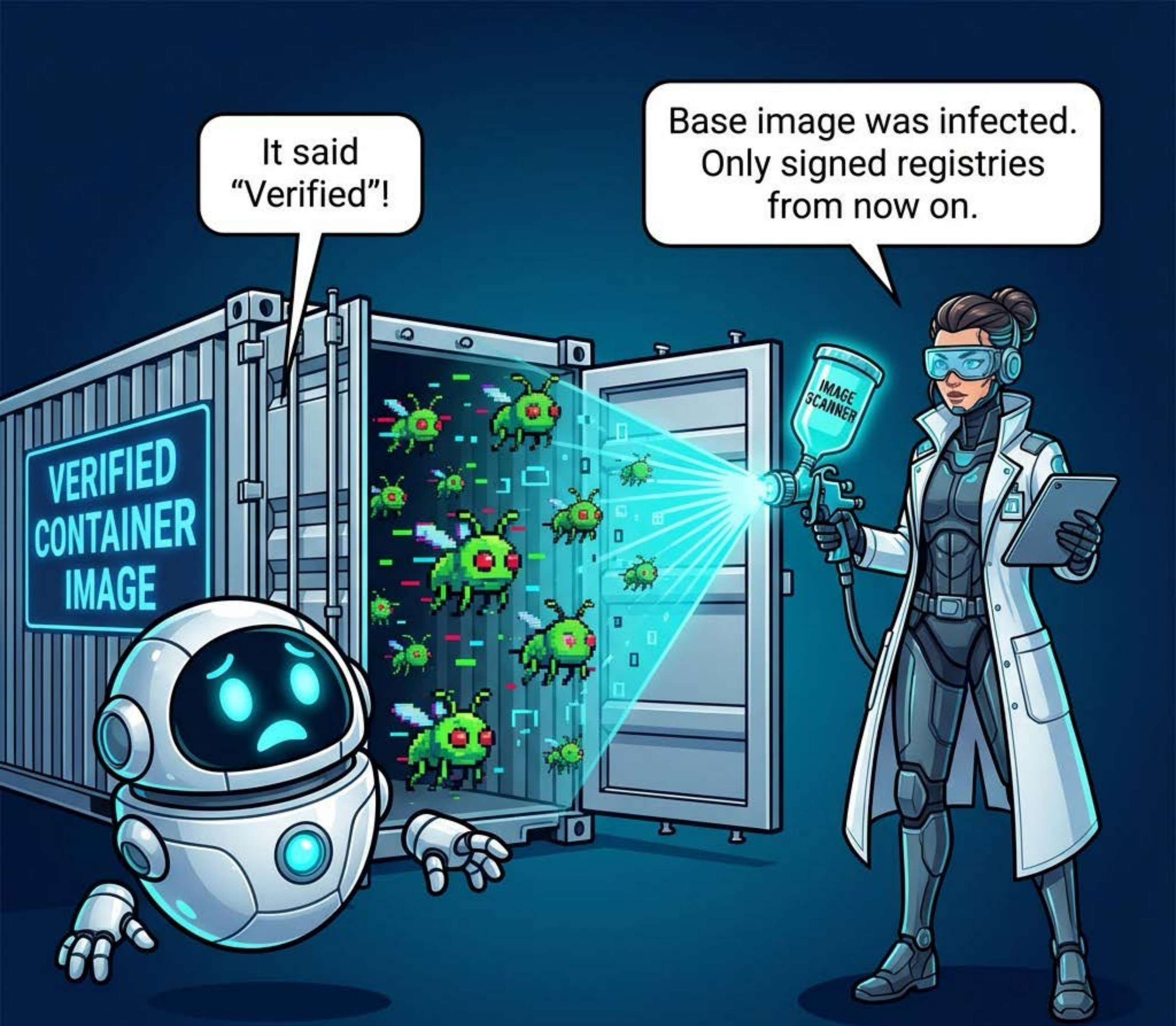
REDACTED

DATA LEAKAGE THROUGH OBSERVABILITY

POISONING OBSERVABILITY DATA

THE 'DECOY ALARM' ANALOGY





COMPROMISED CONTAINER IMAGES:
The ‘Verified’ Trap


Mutiny on the cluster!
RBAC lockdown! No unauthorized captains allowed.
RBAC
KUBERNETES CLUSTER
ORCHESTRATION ATTACKS:
Preventing Unauthorized Control

Is it... airy?

They tampered with the script. GitOps reviews would have caught this.

INFRASTRUCTURE-AS-CODE (IaC) MANIPULATION

PROTECTING DEPLOYMENTS WITH VERSION CONTROL & REVIEW

DoS ATTACK:
DENIAL OF SERVICE



SOLUTION: AUTO-SCALING & FILTERING

RESOURCE HIJACKING

ENERGY DRAIN

I feel so drained...

DENIED

Crypto-miner! Resource quotas will cut him off.

RESOURCE QUOTAS

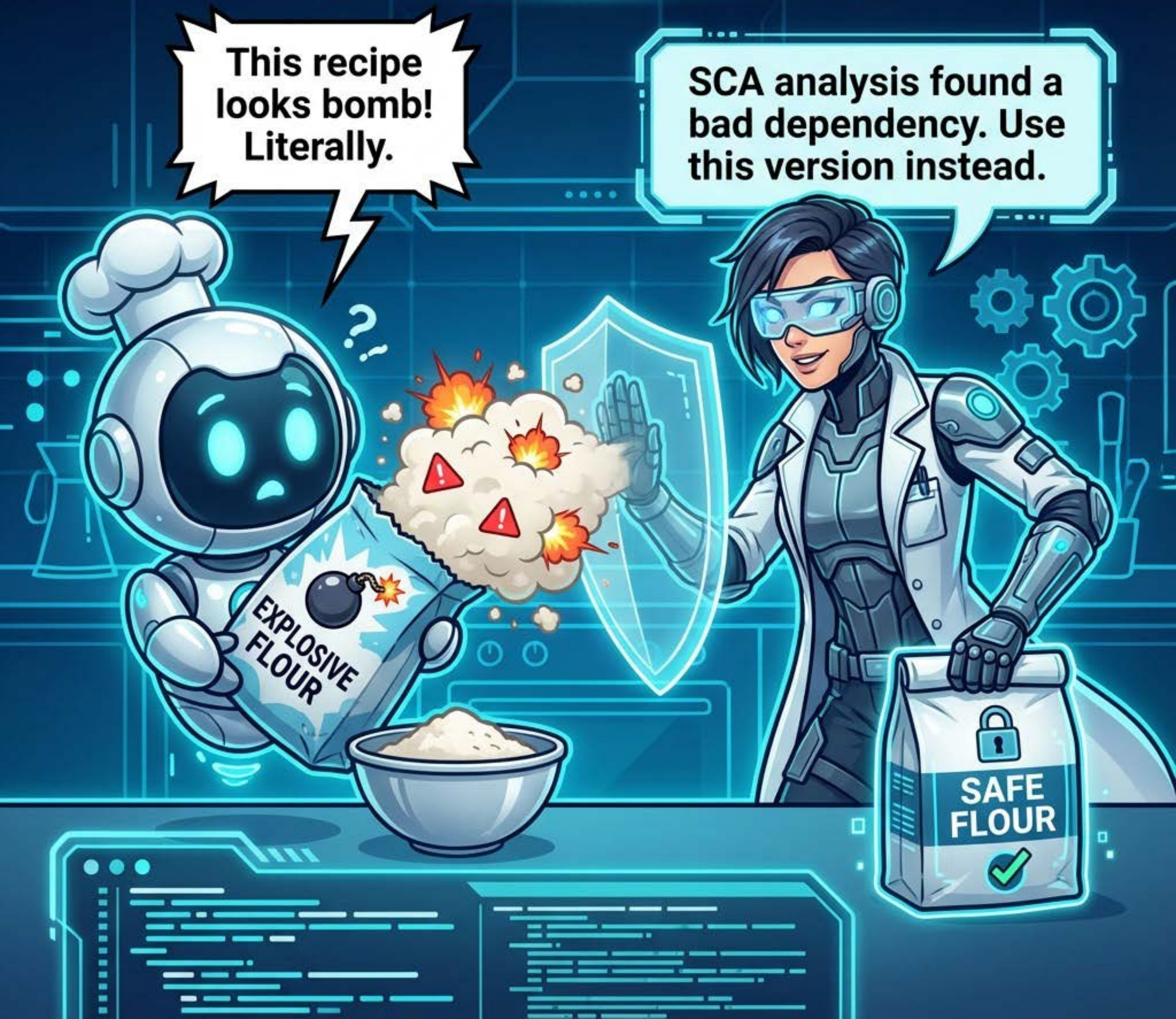
RESOURCE HIJACKING: Unwanted software uses system resources for its own gain.

SOLUTION: Resource quotas limit the amount of energy a process can use.

LATERAL MOVEMENT



COMPROMISED FRAMEWORK COMPONENTS: SCA TO THE RESCUE



I didn't put this here.

Hidden backdoor. Code audits and checksums catch these.

HIDDEN BACKDOOR

CODE AUDITS

CHECKSUMS

TOPIC: BACKDOOR ATTACKS (FRAMEWORK)

INPUT VALIDATION ATTACK!

WHEN MALICIOUS CODE GETS PAST THE GATES!

INPUT: USER DATA ENTERED

VULNERABILITY: LACK OF SANITIZATION

> DELETE ALL

cough

INJECTION ATTACK! SANITIZE YOUR INPUTS, SPARKY!

INPUT: USER DATA ENTERED

VULNERABILITY: LACK OF SANITIZATION

SANITIZE: CLEAN BEFORE PROCESSING!

DANGER

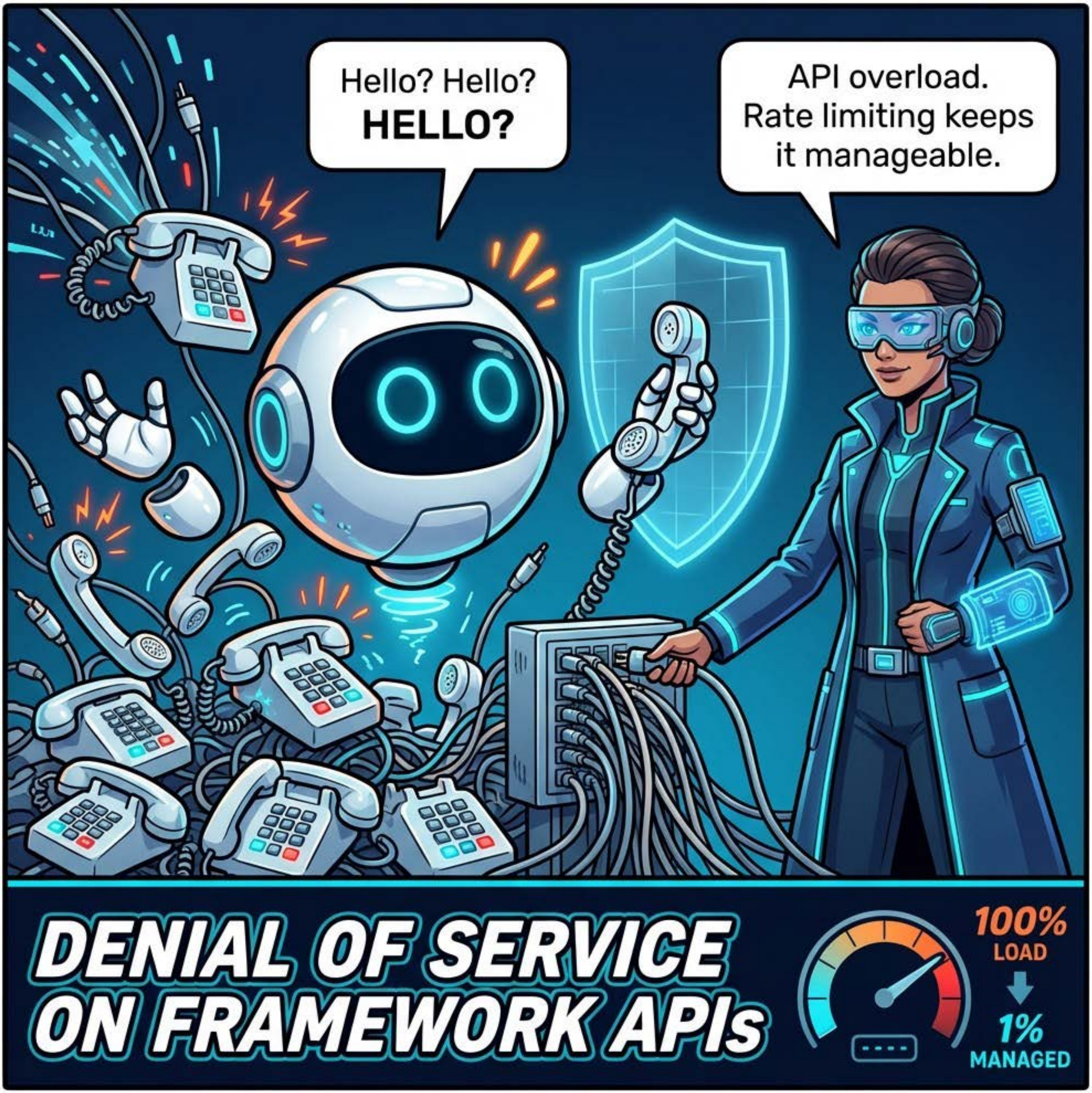
Ahoy!
Just the 'goods',
matey!

HALT!
Verify the
SBOM first.

Look, Dr. Ava!
A new shipment's
here!

SBOM

SUPPLY CHAIN
EXPRESS


Hello? Hello?
HELLO?
API overload.
Rate limiting keeps
it manageable.
DENIAL OF SERVICE
ON FRAMEWORK APIs
100%
LOAD
1%
MANAGED

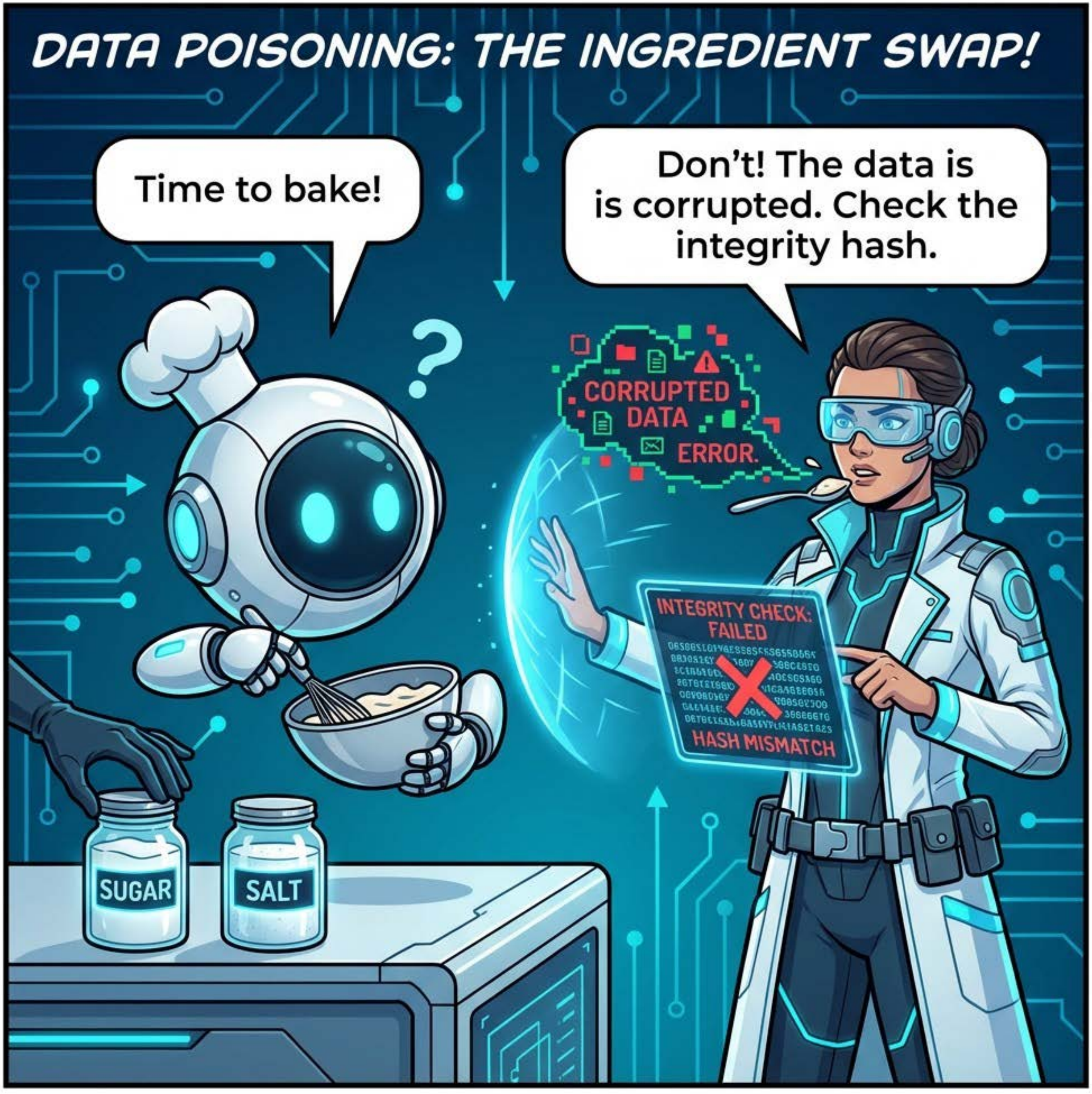
He went under the lock! Framework evasion.

Defense-in-Depth catches what the door misses.

FRAMEWORK EVASION

DEFENSE-IN-DEPTH

DATA POISONING: THE INGREDIENT SWAP!



We're leaking! Data Exfiltration.

Egress filtering stops the flow.

DLP TAPE (Egress Filtering)

DATA LOSS PREVENTION

SECURE DATA FLOW

EGRESS FILTERING ACTIVE

DATA EXFILTRATION & EGRESS FILTERING

Egress filtering monitors and restricts unauthorized data leaving a network, preventing data loss.

DENIAL OF SERVICE ON DATA INFRASTRUCTURE: READ REPLICAS TO THE RESCUE!

DATA LIBRARY

DoS ATTACK

READ REPLICA TUNNEL

DATA LIBRARY

Database DoS. Switch to the Read Replica.

I can't reach the books!

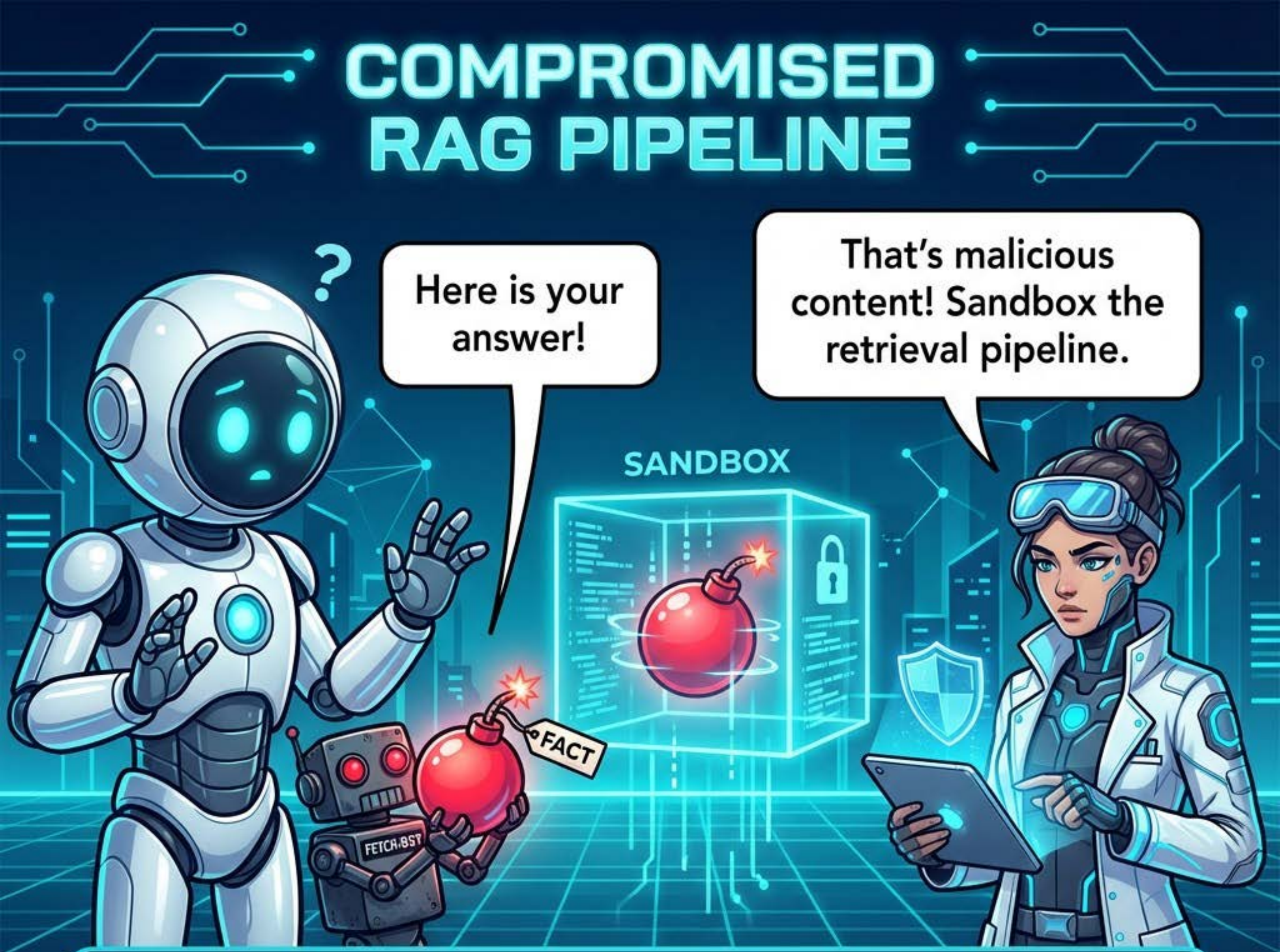
The record is blank.

It was altered. Crypto-hashing proves it's been tampered with.

HASH MISMATCH!

DATA TAMPERING DETECTED

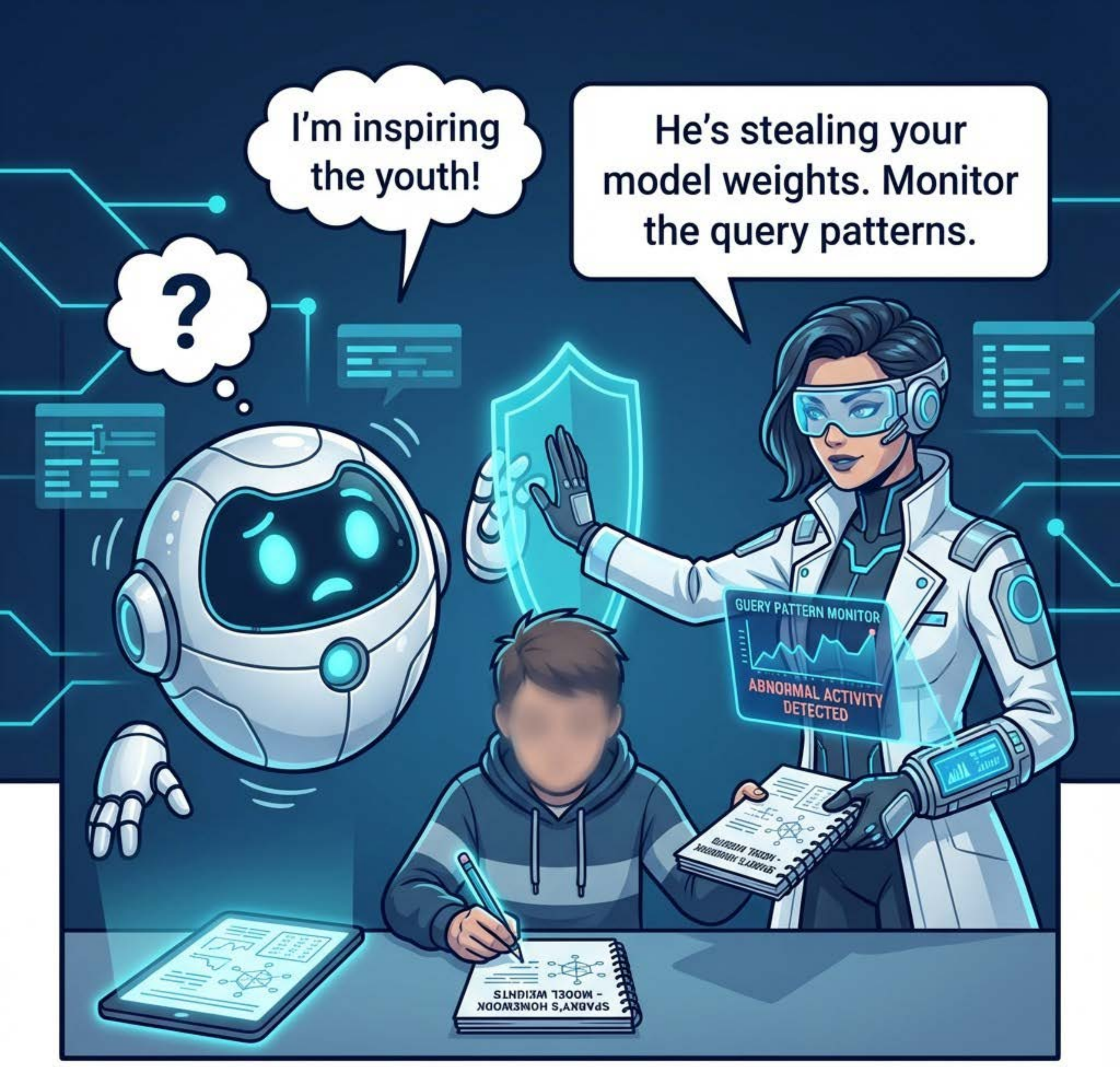
COMPROMISED RAG PIPELINE





ADVERSARIAL EXAMPLES




I'm inspiring the youth!
?
He's stealing your model weights. Monitor the query patterns.
GUERY PATTERN MONITOR
ABNORMAL ACTIVITY DETECTED
SPARKY'S HOMEWORK - MODEL WEIGHTS

BACKDOOR ATTACKS (MODEL)



How did he know that?

MEMBERSHIP INFERENCE ATTACK

Privacy leak. Differential Privacy hides the specifics.

TRAINING DATA: CAT - MATCH FOUND!

DIFFERENTIAL PRIVACY SHIELD

DATA POISONING (TRAINING PHASE)

MALICIOUS DATA

LEARNING FUNNEL

That's garbage. Curate the dataset before training.

DATA CURATION

Yummy knowledge!

?

QUARANTINE

REPROGRAMMING ATTACKS

You're repurposed! Task-specific constraints only.

I'm winning!

MALWARE

SECURITY PATCH

DOOM

Dr. Ava deploys a patch, reorienting Sparky's function.

Sparky is tricked into performing unintended actions.

DENIAL OF SERVICE (MODEL) - SPONGE ATTACK

Preventing Infinite Resource Drain



THE ADAPTIVE SECURITY FRONTIER: PREPARING FOR THE UNKNOWN.

SPARKY

DR. AVA

EMERGING THREATS

MONITORING

EVOLVING DEFENSE

UNKNOWN ANOMALIES

Whatever threats we discussed here, those are not the only threats we have but there are more threats out there we need to deal with them. Stay safe. See you in next part...

SECURE ADAPTIVE FRAMEWORK